

PRODUCTTANK BENGALURU

Platform Thinking in the Real World

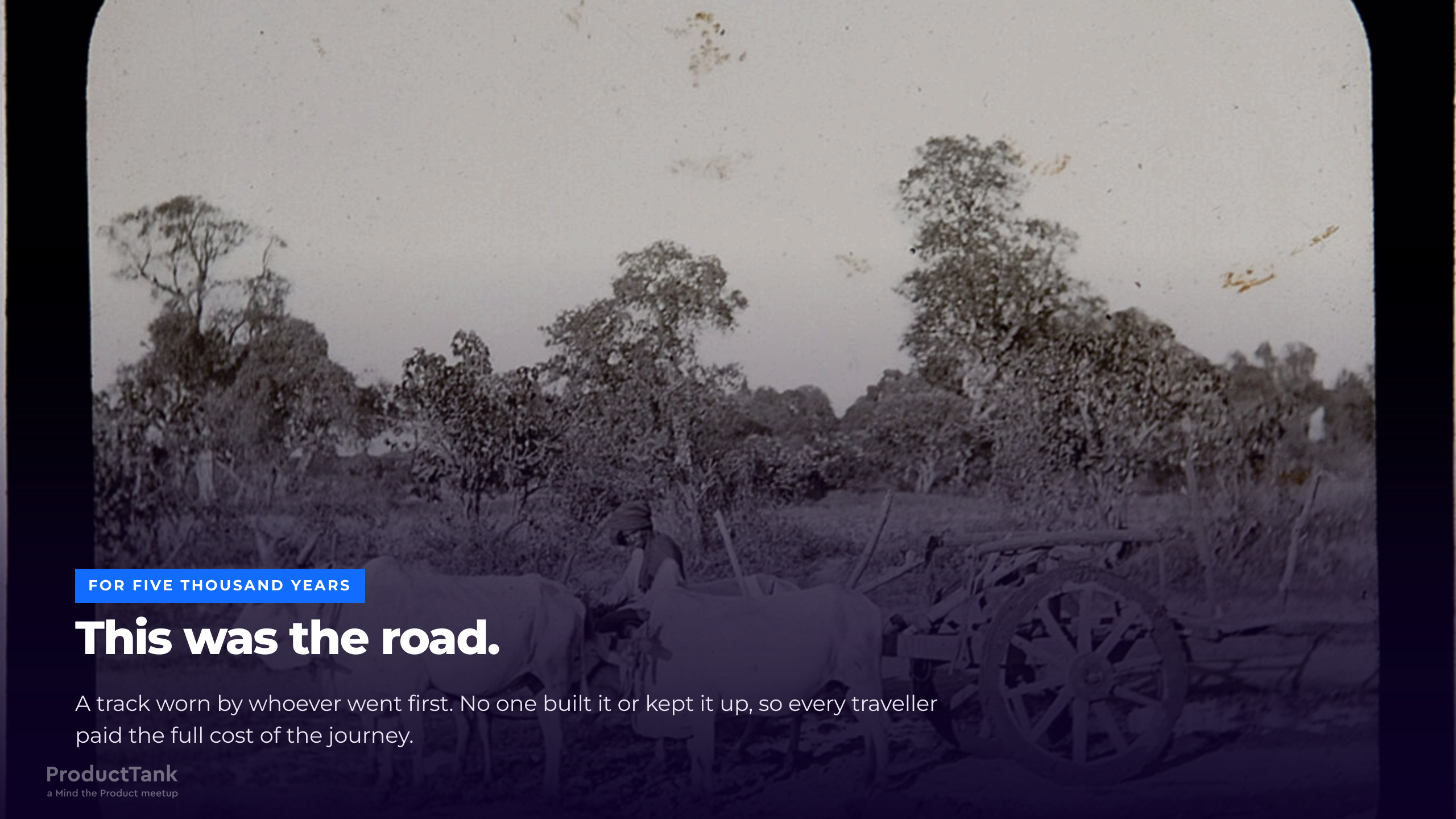
People, process & mindset.

Two thousand three hundred years of road building, and the new driver who arrived on a road somebody else designed.

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FOR FIVE THOUSAND YEARS

This was the road.

A track worn by whoever went first. No one built it or kept it up, so every traveller paid the full cost of the journey.



312 BC

Then somebody built one on purpose.

And the difference lasted two thousand years.

The Appian Way

Construction of the **Via Appia** began in **312 BC**. The Romans engineered the parts you cannot see: **drainage**, a **cambered profile** that sheds water, and **connected routes** that joined other networks.

THE ROAD

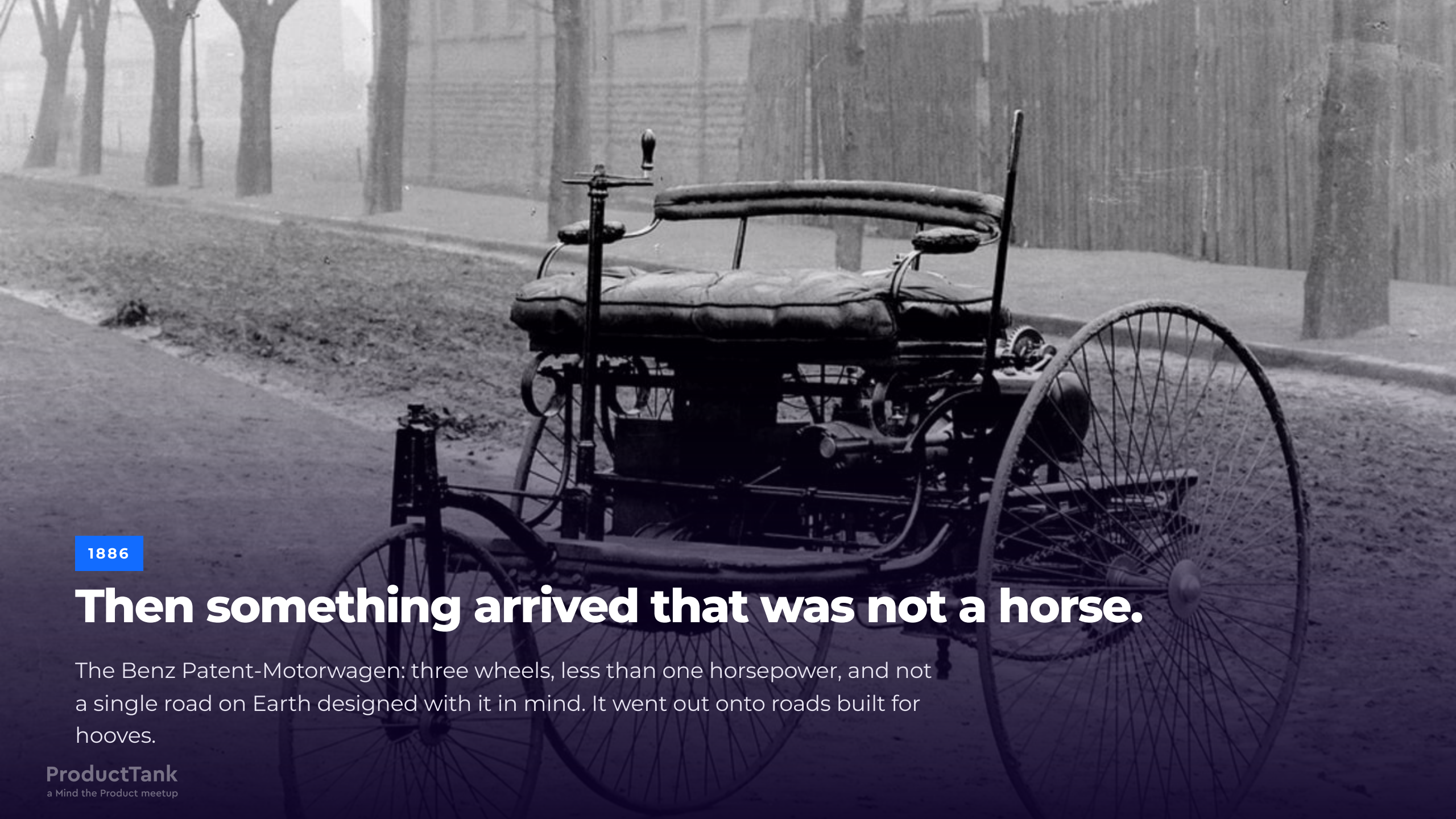
Once somebody builds a road properly, every journey after it becomes cheaper, because the decisions are made once, in the road itself, instead of again by every traveller.



1800S

Two thousand years later.

Engineers kept making roads better and cheaper the whole time, and they built all of them for the same traveller. The fastest way to cross a country was still a horse.



1886

Then something arrived that was not a horse.

The Benz Patent-Motorwagen: three wheels, less than one horsepower, and not a single road on Earth designed with it in mind. It went out onto roads built for hooves.



AUGUST 1888

Then somebody proved it was useful.

Bertha Benz

In **August 1888**, **Bertha Benz** drove from **Mannheim to Pforzheim**, roughly **106 km one way**, with her two sons, and without telling her husband.

She dealt with the problems as they arrived: a blocked fuel line, worn brakes, failing insulation. She fixed them herself, at the roadside, with what she had.

She proved something no patent could: the machine **worked over real distance, in the real world**.

Mannheim → Pforzheim, ~106 km one way, August 1888.



1865

The man with the red flag.

Britain's **Locomotives Act 1865** required a person to walk **at least 60 yards ahead** of certain road locomotives, **carrying a red flag**.

60 yd

ESCORT DISTANCE AHEAD

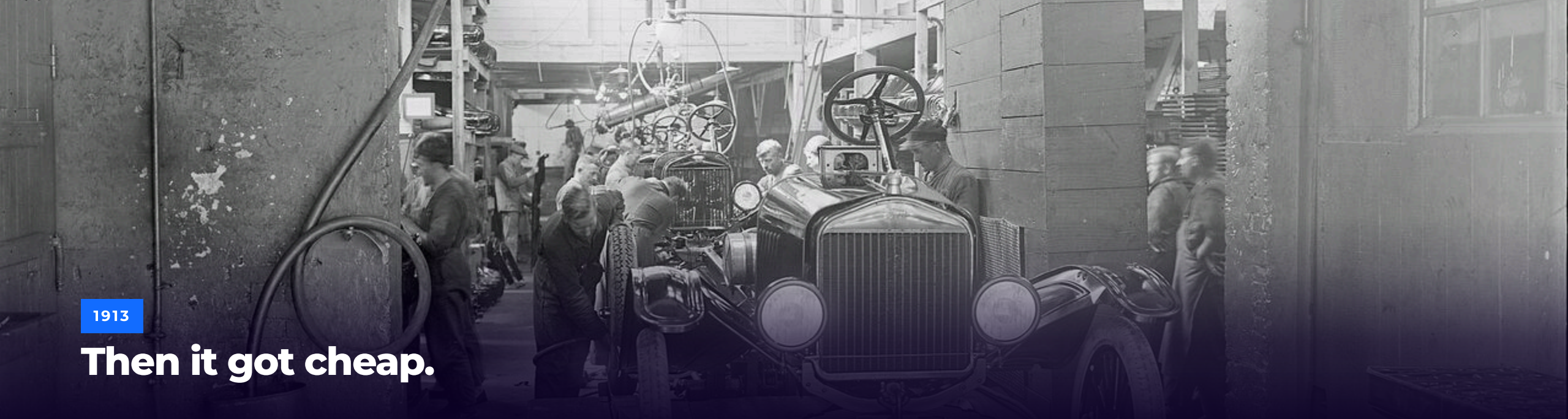
4 / 2 mph

COUNTRY / TOWN LIMIT

1

HUMAN PER MACHINE

Picture a powerful new machine with a person walking in front of it, waving a flag.



1913

Then it got cheap.

The scale curve

Registered motor vehicles in the United States:

8,000

1900

~10M

1920

26.7M

1930

Somewhere in the middle of that curve, everybody sorting it out for themselves stops working. Nobody did anything wrong. The cost of everybody agreeing separately grows faster than the traffic does.

Source: FHWA Highway Statistics historical series (MV-200).



THE 1900S

This was the coordination mechanism.

One man, one junction, making every decision fresh for as long as he stood there.



THE 1920S

And everything broke.

Eleven signs for one route

Early automobile clubs put up their own route signs. Each club had its own scheme, and none of them coordinated.

The Federal Highway Administration's own history records that on some heavily travelled routes, a driver could encounter **as many as eleven different signs for a single route.**

Eleven signs. One road. **Every one of them technically correct.**

Source: FHWA highway history.

Then, in twelve years

The fix was already on the road: a **centreline**, painted in Michigan a decade earlier, credited to **Edward N. Hines**.

1911**Centreline**

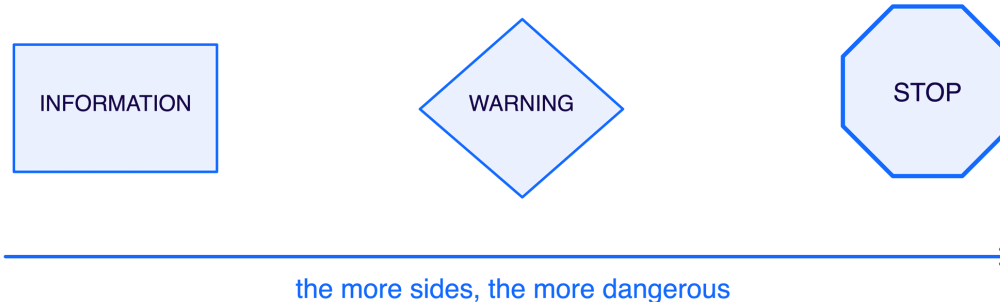
Painted road marking, Michigan.

1914 → 1920**Traffic signals**

First permanent electric signal, Cleveland.
Then the tri-colour signal, Detroit, credited to William Potts.

1923**Shape = meaning**

Sign shapes standardised, so meaning is readable before the text is.



Shape encodes severity: the octagon reads as a stop before you read the word.

THE ORDER

Build the road first. Encode the rules second. Ask the driver to be competent third. Britain's 1903 licences identified the driver and tested nothing.

Mississippi Valley Association of State Highway Departments, 1923. AASHO's combining report followed in 1924; its manual in 1927. First MUTCD, 1935 (§1A.06).



1930S

Then we rebuilt the road around the machine.

Grade separation. Controlled access. Curves designed for a speed nobody could have reached in 1886.



1956

And did it at continental scale.

The Interstate programme

Congress created the **mandate** for **uniform geometric and construction standards** in the **Federal-Aid Highway Act of 1956**, and **AASHO**, the engineering body, wrote the **specification** the federal highway authority adopted. Neither could have done the other's job.

1956, DESIGNING FOR 1975

The standards specified a capacity horizon: engineers sized the roads for the traffic expected in 1975, traffic that would not exist for twenty years. And they were right.

Federal-Aid Highway Act of 1956; AASHO design standards; FHWA on the twenty-year design period.

And it worked

Interstates carry traffic far faster than the roads they replaced. If rules and barriers were a tax on speed, you would expect them to be more dangerous.

0.85

INTERSTATE FATALITIES PER 100M VEHICLE-
MILES

1.53

OTHER ROADS, SAME MEASURE

1.8×

THE RISK YOU AVOID BY DRIVING THE BUILT
ROAD

Faster roads. Roughly half the fatality rate. The guardrails made the speed survivable.

Source: FHWA. The 1.8× is the ratio of the two published FHWA figures.

TODAY

Two thousand three hundred years, finished.

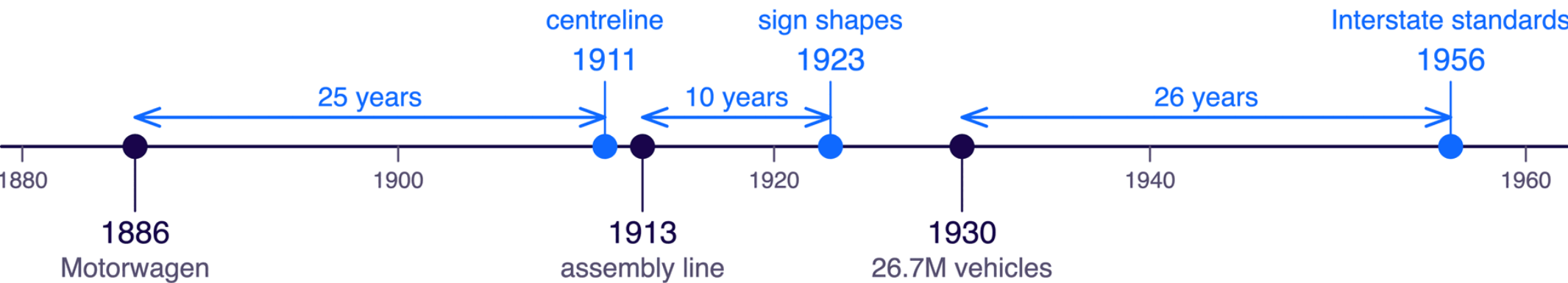
No driver out there is thinking about drainage, or sign shapes, or who decided which side to drive on. That is the point.

The obvious conclusion is wrong

Two thousand three hundred years of that, and the tidy conclusion is that the road comes **first**.

That is not what you just watched.

the road caught up



the machine arrived

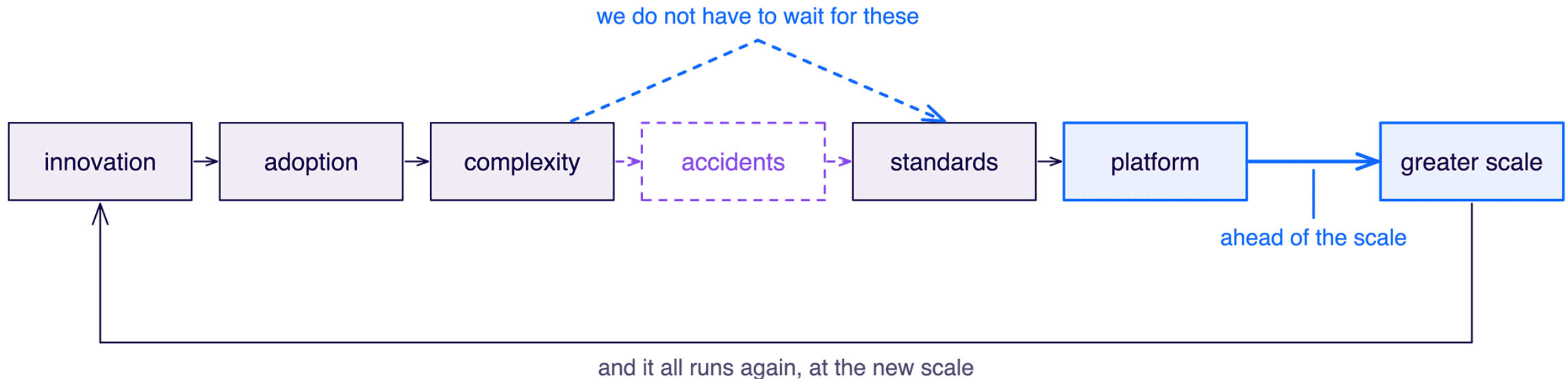
The machine arrived first every time, by decades.

Ahead of the scale

THE THESIS

The platform does not have to be ahead of the innovation. It has to be ahead of the scale.

The cycle the road record documents:



Innovation arriving first is normal. Arriving at **26 million vehicles** still running 1900's coordination model is the failure.

CASE STUDY

Then it was my system, behind the scale



Homecare software, fifteen years old, same lag the roads had.

A fifteen-year-old product, and an uncomfortable diagnosis

Homecare: care delivered in the patient's home (nursing, therapy, personal care, remote monitoring) by clinicians in the field, on phones, with unreliable connectivity. Every record is **PHI**, so **there is no "just ship it" path**.

I was leading engineering for the product. Four things were degrading at once: **stability, security, compliance, delivery velocity**. Finance saw four bids for budget; the roadmap saw every estimate quietly growing. They were one problem wearing four costumes.

NO SEPARATION OF CONCERNS

Domain logic, persistence, compliance behaviour and UI reasoning interleaved. You could not change one without auditing all four.

NO PLATFORM FOUNDATION

Every team re-solved identity, audit trails, permissions, forms and data shape, each in its own way.

THE SYSTEM OF RECORD WAS OURS

A bespoke schema meant every new customer concept became a modelling project before it became a feature.

THE DIAGNOSIS

Stability, security, compliance and velocity degrade together, because they share one root cause: the absence of boundaries.

The bet: stop paying to invent our own system of record

FHIR, HL7's Fast Healthcare Interoperability Resources, gave us a schema we could have written ourselves and an ecosystem we could not have.

THE METADATA ALREADY EXISTS

Patient, Encounter, Observation, CarePlan, Practitioner. It already modelled every healthcare concept our customers could name.

FORMS COME WITH IT

Questionnaire and QuestionnaireResponse, plus the Structured Data Capture guide. LOINC supplies the standardised instruments.

YOU DO NOT HAVE TO HOST IT

Managed FHIR backends exist on all three hyperscalers. Mature self-hosted open source exists too, if you would rather own it.

We stopped paving our own road and drove on somebody else's: one a global community already maintains, tests and extends. That was a build-or-buy decision before it was an architecture decision: not **is the standard perfect**, but **is our own schema worth what we pay, every release, to keep it alive**.

Managed: Azure Health Data Services, AWS HealthLake, Google Cloud Healthcare API. Open source: HAPI FHIR, Medplum, LinuxForHealth FHIR Server.

The pilot: three months, five fresh graduates

We skipped the strategy document and built a narrower version of our own product.

Month 1 Foundations ABP.io for the application platform: modular, multi-tenant, with audit logging and permission management built in. FHIR for the system of record.	Month 2 A working lite product Narrower than what we sold, and it ran.	Month 3 Enterprise-grade, audit-ready Delivered by five engineers who had graduated that year.
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Five graduates. Three months. Audit-ready. Who achieved it mattered more than how fast. If the platform makes new graduates dangerous, the platform is real, and your delivery estimates stop depending on which five people happen to be free.

The part that made it work

The technology was the easy half. The other half decided the outcome.

01

WE PUT THEM ON THE FLOOR

The team sat in the middle of the floor, not in a side room. They built a startup atmosphere there, and people walking past caught it.

02

WE SOCIALISED IT IN THE OPEN

On the floor, and one-to-one. We pulled people into the discussion as **advisors** (including the stakeholders who could have blocked it), asking for their help and their opinions rather than their sign-off.

03

THEY MADE IT THEIRS

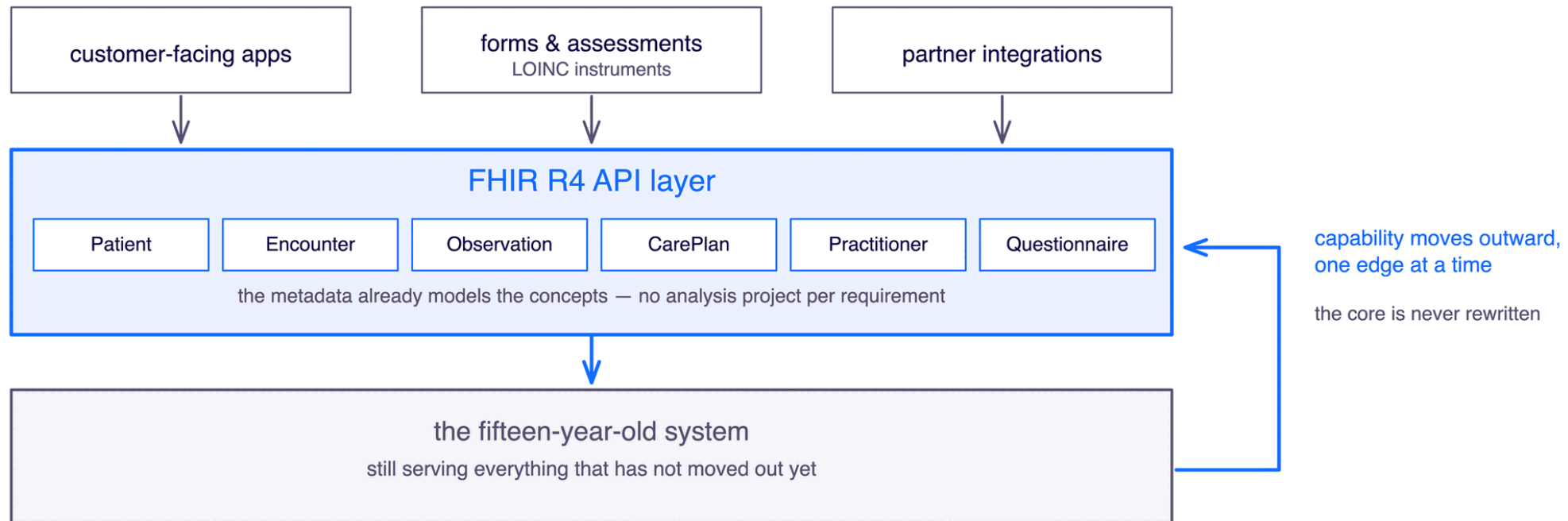
By the end of the three months everybody was contributing and defending it. The vision that had looked impossible now looked reachable.

ADOPTION

People adopt the platform they were consulted on. Ask for advice long before you ask for adoption.

Then we did it to the real product

The pilot was greenfield. The fifteen-year-old system was not. So instead of the big-bang rewrite you keep being asked to fund, we put a **FHIR R4 API layer in front of it**: a strangler fig.



Within **two months**: business analysis stopped being a project, forms stopped being a backlog, and the sceptics converted themselves. The second pilot ran inside the product they already worked on every day.

Strangler fig application, Martin Fowler. The pattern replaces a legacy system incrementally, at the edges, without a big-bang rewrite.

Then the org chart stopped matching the system

We had named our teams after **activities**: enhancement, tech debt, UI revamp. Each one re-solved the same foundations its own way. The platform layer gave us boundaries the org chart did not have, so we redrew the teams around them.



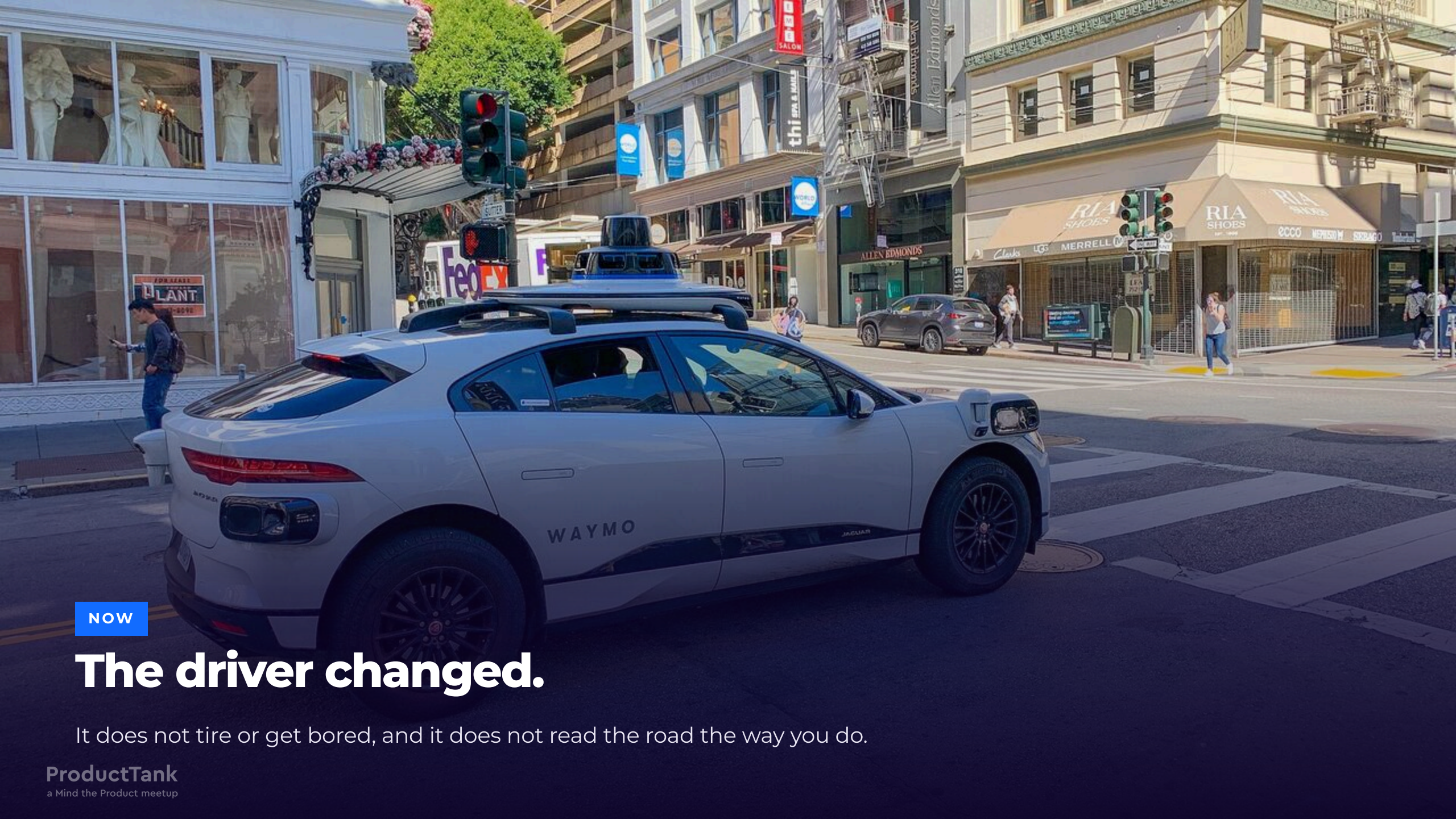
The band is the widest thing on the chart, and **the team that owns it is two people**. Every team above it used to keep its own copy of everything inside it.

Conway's law (Melvin Conway, 1968): organisations produce designs that copy their own communication structures. Changing the org to get the architecture is the inverse Conway manoeuvre.

We stopped re-deciding problems that were already solved.

That is where the speed came from. Five graduates proved it before we asked anyone to believe it. Decisions that live in the environment do not care who is driving.

What happens when the driver changes?



NOW

The driver changed.

It does not tire or get bored, and it does not read the road the way you do.

For over a century, we optimised around one assumption

TRANSPORTATION

Signage legibility, reaction-time margins, lane widths, signal timing, licensing: engineers calibrated each one around the capabilities and limits of a **human driver**.

PRODUCT AND SOFTWARE ENGINEERING

Spec review, code review, roadmap cadence, approval gates, onboarding docs: we calibrated each one around the capabilities and limits of a **human team**: people who decide at meeting speed and ship at review speed.

Both platforms are **human-scaled**. We shaped them to the operator. In software, that operator has just changed: **AI agents** now write, review, deploy and operate the system, and draft the specs and tickets that describe it.

This already happened once.

In **1886**, a far more capable participant arrived on infrastructure designed for the previous generation of participant.

The infrastructure was **correctly designed for the wrong user**.

Five assumptions about the driver

Our platform is human-scaled: the process as much as the pipeline. Each of these gates assumes the operator is a person.

Spec review and code review assume a human bottleneck is acceptable. We made the slowest step slow on purpose.

Quarterly roadmaps assume change arrives at human frequency.

Estimates assume throughput is a function of how many people you have.

Approval gates and staging environments assume somebody is watching when the change lands.

Documentation and "ask Amit" assume a reader who asks a colleague when it is unclear, and that the participant can, in fact, ask Amit.

Each of these made sense when we built it. **All of them are calibrated for the previous driver.**

When the driver becomes less predictable, the road must become more predictable.

Meetings will not contain a probabilistic operator. Move the decisions **out** of human review and **into** the environment, where they stay deterministic, checkable and always on.

For AI agents, we are somewhere around 1905.

The participant has arrived and adoption is compounding, while the coordination model is still the previous generation's.

We know what comes next in this sequence. **We do not have to wait for the accidents to write the standards.**

CLOSE

What to do on Monday

Five questions for your platform and your process

1. COUNT YOUR SIGNS

How many ways can a decision get made here, and how many ways can a change reach production? If either answer is more than two, you have the 1920s problem.

2. FIND THE RED FLAGS

Which of your gates (approvals, sign-offs, review meetings) exist because a human used to be the slowest part? Is that still true?

3. NAME YOUR CAPACITY HORIZON

1956 designed for 1975. What load is your product built for, and do your roadmap and your platform name the same number?

4. FIND YOUR 106 KM

Bertha Benz settled the argument by driving the thing, not by describing it. What is your existence proof, and who is refusing to fund you until they see one?

5. PAINT ONE LINE

Pick the single decision your org re-negotiates most (the one argued in every planning meeting) and encode it once, in the environment, this quarter.

Thank you

Questions, arguments and corrections all welcome, especially the corrections. I had to make nine of them to this talk already.

Slides github.com/v1bh0r/pt-bengaluru-talk-aug-2026

Sources Every claim is dated and sourced: [full bibliography](#)

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Vibhor Mahajan

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